Computer Graphic

B. Sc.CSIT and BIT [TU]

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Unit 1: Introduction and Overview of Graphics System (3 Hrs.)

- Definition and Representative uses of Computer Graphics, Computer Graphics vs. Image Processing,
- Application Areas, Overview of Coordinate System, Definition of Scan Conversion,
- Rasterization and Rendering, Raster Scan & Random Scan Displays, Architecture of Raster
- Graphics System with Display Processor, Architecture of Random Scan Systems

❑ Definition and Representative uses of Computer Graphics, Computer Graphics vs. Image Processing,

- Graphics are defined as any sketch or a drawing or a special network that pictorially represents some meaningful information.
- Computer Graphics is used where a set of images needs to be manipulated or the creation of the image in the form of pixels and is drawn on the computer.
- Computer Graphics can be used in digital photography, film, entertainment, electronic gadgets, and all other core technologies which are required.
- It is a vast subject and area in the field of computer science.
- Computer Graphics can be used in UI design, rendering, geometric objects, animation, and many more.
- In most areas, computer graphics is an abbreviation of CG.

Computer graphics is an art of drawing pictures on computer screens with the help of programming. It involves computations, creation, and manipulation of data. In other words, we can say that computer graphics is a rendering tool for the generation and manipulation of images.

❑ Computer Graphics vs. Image Processing

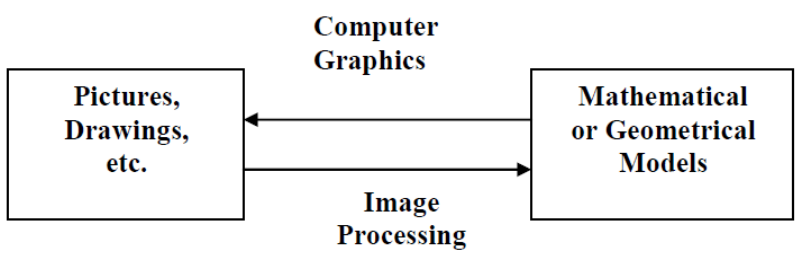
Image processing and computer graphics are two separate but closely related fields of digital signal processing. Image processing is the manipulation of digital images through an algorithm, while computer graphics are the creation of images through the use of computer programs.

Image processing algorithms are used in order to improve the quality of digital images, or to extract specific information from them.

For example, image processing can be used to remove noise from an image, or to sharpen it.

Computer graphics, on the other hand, are used to generate images from scratch, or to modify existing images.

Both image processing and computer graphics make use of digital signal processing in order to achieve their goals. However, the two fields are distinct, and each has its own unique set of challenges and opportunities.



- ❑ Computer Graphics: Synthesize pictures from mathematical or geometrical models.
- ❑ Image Processing: analyze pictures to derive descriptions (often in mathematical or geometrical forms) of objects appeared in the pictures.

❑ Types of Computer Graphics

1. Raster Graphics: In raster, graphics pixels are used for an image to be drawn. It is also known as a bitmap image in which a sequence of images is into smaller pixels. Basically, a bitmap indicates a large number of pixels together.
2. Vector Graphics: In vector graphics, mathematical formulae are used to draw different types of shapes, lines, objects, and so on.

❑ Applications

- Computer Graphics are used for an aided design for engineering and architectural system- These are used in electrical automobiles, electro-mechanical, mechanical, electronic devices. For example gears and bolts.
- Computer Art – MS Paint.
- Presentation Graphics – It is used to summarize financial statistical scientific or economic data. For example- Bar chart, Line chart.
- Entertainment- It is used in motion pictures, music videos, television gaming.
- Education and training- It is used to understand the operations of complex systems. It is also used for specialized system such for framing for captains, pilots and so on.
- Visualization- To study trends and patterns. For example- Analyzing satellite photo of earth.

❑ Cathode Ray Tube

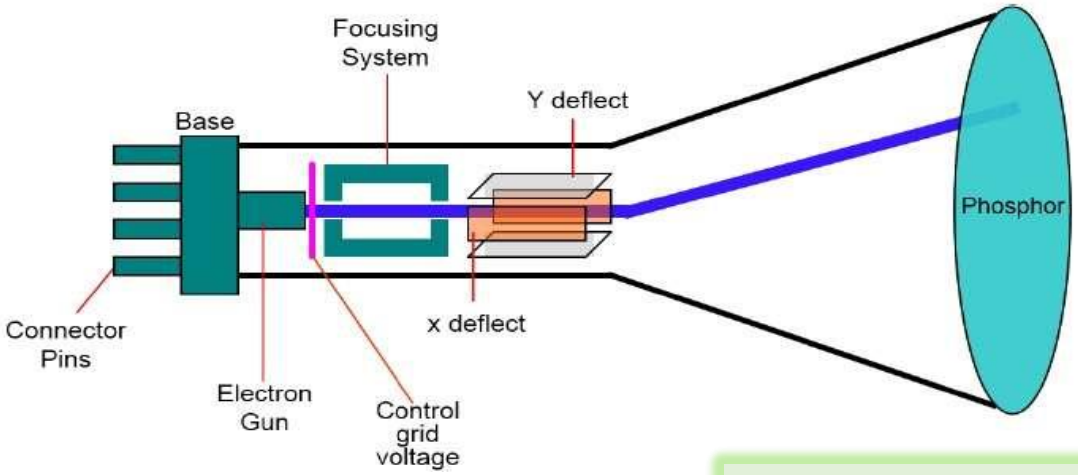
1. Electron Gun: Electron gun consisting of a series of elements, primarily a heating filament (heater) and a cathode. The electron gun creates a source of electrons which are focused into a narrow beam directed at the face of the CRT.

2. Control Electrode: It is used to turn the electron beam on and off.

3. Focusing system: It is used to create a clear picture by focusing the electrons into a narrow beam.

4. Deflection Yoke: It is used to control the direction of the electron beam. It creates an electric or magnetic field which will bend the electron beam as it passes through the area. In a conventional CRT, the yoke is linked to a sweep or scan generator. The deflection yoke which is connected to the sweep generator creates a fluctuating electric or magnetic potential.

5. Phosphorus-coated screen: The inside front surface of every CRT is coated with phosphors. Phosphors glow when a high-energy electron beam hits them. Phosphorescence is the term used to characterize the light given off by a phosphor after it has been exposed to an electron beam.

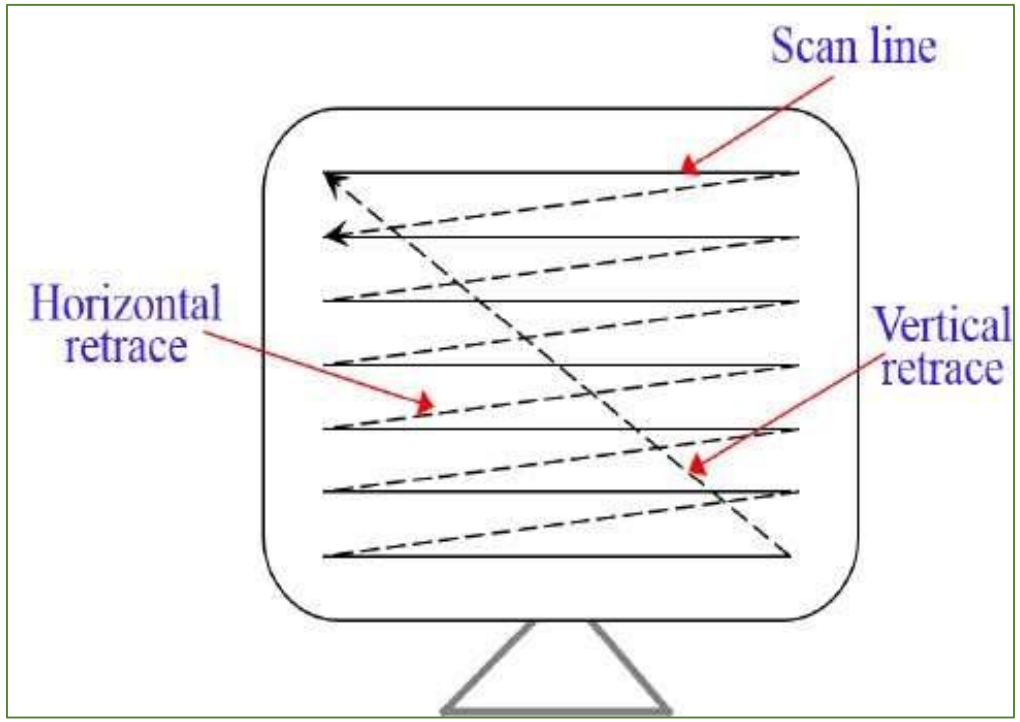


There are two ways Random scan and Raster scan by which we can display an object on the screen.

❑ Random Scan and Raster Scan Display:

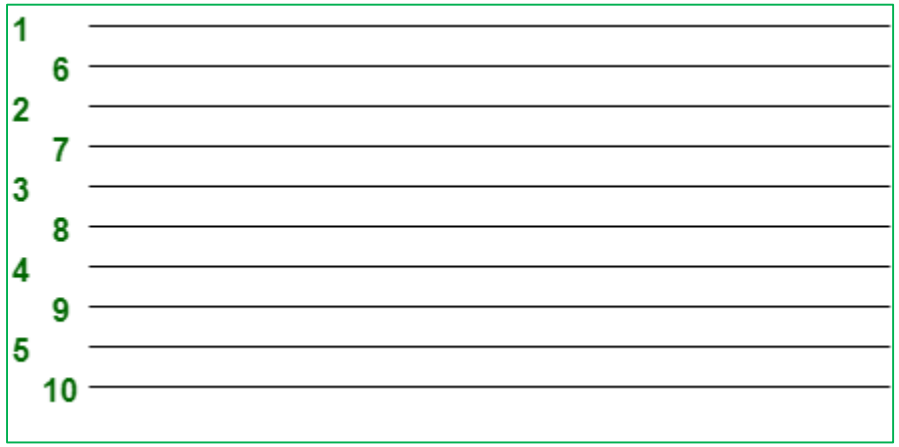
Basic working of Raster Scan

- In this system, a beam of an electron is moved across the screen. It moves from top to bottom considering one row at a time.
- As the beam of electron moves through each row, its intensity is alternatively turned on and off which helps to create a pattern of spots that are illuminated.
- When each scan of the line is refreshed it returns to the left side of the screen. This motion is known as Horizontal retrace.
- As a particular frame ends, the beam of electron moves to the left top corner of the screen to move to another frame. This motion is referred to as Vertical retrace.
- The picture is then stored in an area of memory which is referred to as the frame buffer or refresh buffer.
- The buffer in a raster scan is that area that is responsible for containing intensity of the various points on the screen.
- The values stored in the buffer are then fetched and traced over scan lines one by one on the screen
- The image formed through this raster scan is known as a raster image. The quality of this image is determined by the number of pixels which is termed as the **resolution of the image**.
- The amount of information each pixel represents is known as the **color depth of the image**.
- The raster graphics system of high quality contains 24 bits per pixel in the frame buffer.
- This is referred to as a **full color** or **true color** system. Refreshing of raster scan displays is carried out at the rate of 60 to 80 frames per second.



❖ Interlacing

A TV video signal is graphically interlaced, which means every full screen of information is made up of two separate fields which include the odd field and even field. First, the odd lines are printed on the graphics screen. Then, the even lines are printed in between the odd lines before the odd lines fade away. This all happens faster than any human eye can perceive.



For non-interlaced display refresh rate of 30 frames per second used. But it gives flickers. For interlaced display refresh rate of 60 frames per second is used.

Random Scan Display

- In Random Scan Display a beam of the electron is directed only to the screen areas where any picture has to be displayed or drawn on the screen.
- It is also termed as vector display, as it displays or draws a picture in the form of one line at a time.
- It can draw and refresh lines on the screen of a picture in any sequence not particularly specific.

Basic working of random scan display

- Random scan monitors are used to draw a picture in one line at a time and are thus also referred to as **vector displays**.
- The cathode ray tube when operates as a random scan display device directs the beam of an electron only to those areas of the screen where display or a picture has to be drawn.
- To draw a picture or display it on the screen the system goes through a line or set of commands and draws each of them one at a time in a line turn by turn.
- The **refresh rate** here depends on the number of lines that are to be displayed on the screen and are designed so that they draw the component lines of the picture 30 to 60 times in a second.
- They have a high resolution of pictures and produce smooth line drawing. It's that smooth that while zooming also it doesn't spread.

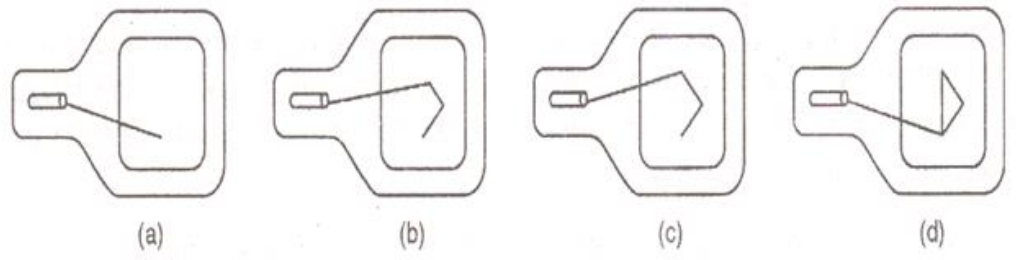


Fig: Random Scan Display

Random Scan Display:

Advantages:

- A CRT has the electron beam directed only to the parts of the screen where an image is to be drawn.
- Produce smooth line drawings.
- High Resolution

Disadvantages:

- Random-Scan monitors cannot display realistic shades scenes.

Raster Scan Display:

Advantages:

- Realistic image
- Million Different colors to be generated
- Shadow Scenes are possible.

Disadvantages:

- Low Resolution
- Expensive

❑ Differentiate between Random and Raster Scan Display:

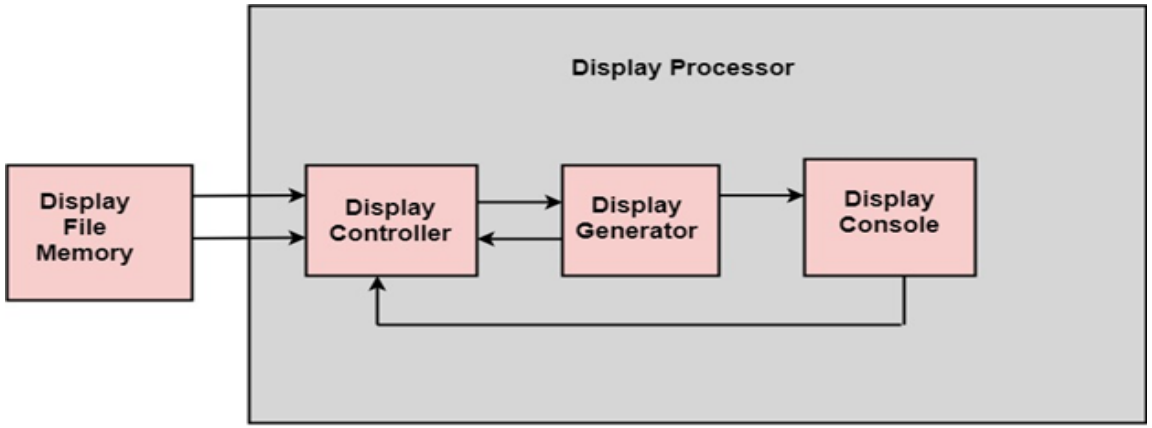
Random Scan	Raster Scan
1. It has high Resolution	1. Its resolution is low.
2. It is more expensive	2. It is less expensive
3. Any modification if needed is easy	3.Modification is tough
4. Solid pattern is tough to fill	4.Solid pattern is easy to fill
5. Refresh rate depends or resolution	5. Refresh rate does not depend on the picture.
6. Only screen with view on an area is displayed.	6. Whole screen is scanned.
7. Beam Penetration technology come under it.	7. Shadow mark technology came under this.
8. It does not use interlacing method.	8. It uses interlacing
9. It is restricted to line drawing applications	9. It is suitable for realistic display

❑ Display Processor:

It is interpreter or piece of hardware that converts display processor code into pictures. It is one of the four main parts of the display processor

Parts of Display Processor

1. Display File Memory
2. Display Processor
3. Display Generator
4. Display Console



Block diagram of Display System

Display File Memory: It is used for generation of the picture. It is used for identification of graphic entities.

Display Controller:
It handles interrupt
It maintains timings
It is used for interpretation of instruction.

Display Generator:
It is used for the generation of character.
It is used for the generation of curves.

Display Console: It contains CRT, Light Pen, and Keyboard and deflection system.
The raster scan system is a combination of some processing units. It consists of the control processing unit (CPU) and a particular processor called a display controller. Display Controller controls the operation of the display device. It is also called a video controller.

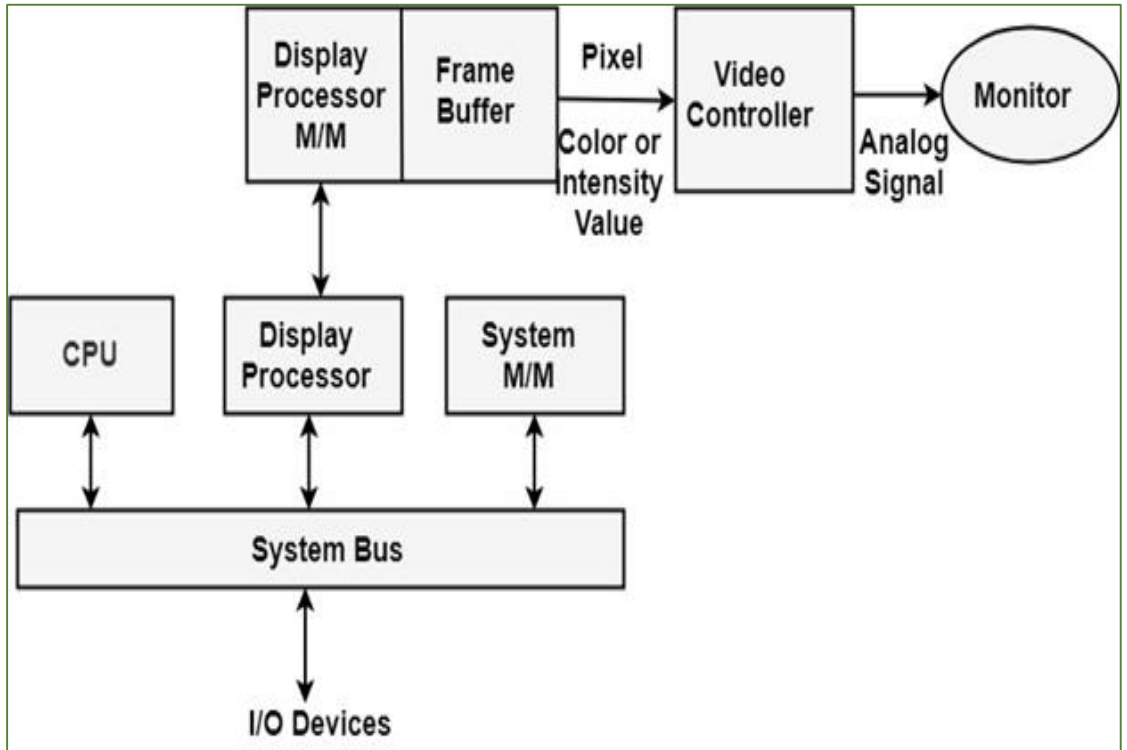
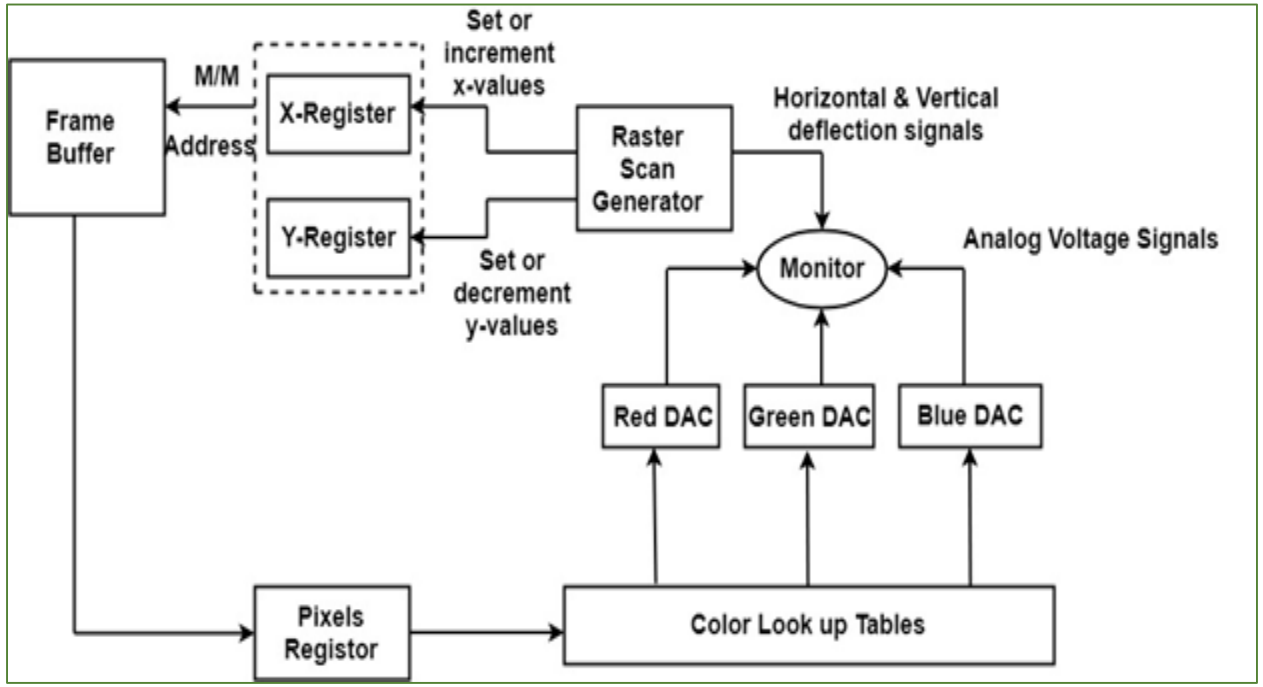


Fig: Architecture of a Raster Display System with a Display Processor



Working:

The video controller in the output circuitry generates the horizontal and vertical drive signals so that the monitor can sweep. Its beam across the screen during raster scans.

As fig showing that 2 registers (X register and Y register) are used to store the coordinate of the screen pixels. Assume that y values of the adjacent scan lines increased by 1 in an upward direction starting from 0 at the bottom of the screen to y_{max} at the top and along each scan line the screen pixel positions or x values are incremented by 1 from 0 at the leftmost position to x_{max} at the rightmost position.

The origin is at the lowest left corner of the screen as in a standard Cartesian coordinate system.

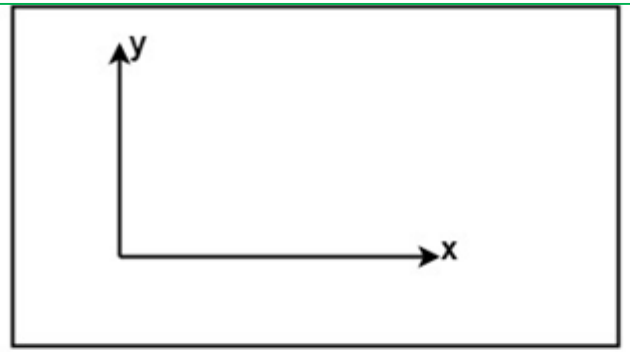


Fig:The origin of the coordinate system for identifying screen positions is usually specified in the lower-left corner.

At the start of a Refresh Cycle:

X register is set to 0 and y register is set to ymax. This (x, y') address is translated into a memory address of frame buffer where the color value for this pixel position is stored.

The controller receives this color value (a binary no) from the frame buffer, breaks it up into three parts and sends each element to a separate Digital-to-Analog Converter (DAC).

These voltages, in turn, controls the intensity of 3 e-beam that are focused at the (x, y) screen position by the horizontal and vertical drive signals.

This process is repeated for each pixel along the top scan line, each time incrementing the X register by Y.

As pixels on the first scan line are generated, the X register is incremented throughxmax.

Then x register is reset to 0, and y register is decremented by 1 to access the next scan line.

Pixel along each scan line is then processed, and the procedure is repeated for each successive scan line units pixels on the last scan line (y=0) are generated.

For a display system employing a color look-up table frame buffer value is not directly used to control the CRT beam intensity.

It is used as an index to find the three pixel-color value from the look-up table. This lookup operation is done for each pixel on every display cycle.

As the time available to display or refresh a single pixel in the screen is too less, accessing the frame buffer every time for reading each pixel intensity value would consume more time what is allowed:

Multiple adjacent pixel values are fetched to the frame buffer in single access and stored in the register.

After every allowable time gap, the one-pixel value is shifted out from the register to control the warm intensity for that pixel.

The procedure is repeated with the next block of pixels,and so on, thus the whole group of pixels will be processed.

Best of Luck